1. 输出方程：$CO = Q_3$

驱动方程：

$$\begin{cases} J_1 = \overline{Q_3} \\ K_1 = \overline{Q_3} \\ J_2 = Q_1 \\ K_2 = Q_1 \\ J_3 = Q_1Q_2 \\ K_3 = Q_1Q_2 \end{cases}$$

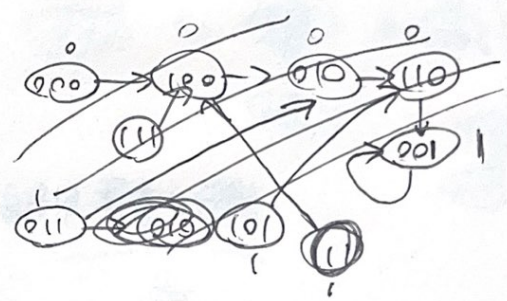
状态方程：

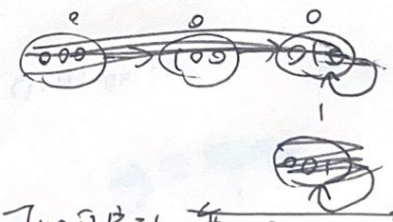
$$Q_1^{n+1} = J_1\overline{Q_1^n} + \overline{K_1}Q_1^n = \overline{Q_3}\,\overline{Q_1^n} + \overline{Q_3}Q_1^n$$

$$Q_2^{n+1} = J_2\overline{Q_2^n} + \overline{K_2}Q_2^n = Q_1^n\overline{Q_2^n} + \overline{Q_1}Q_2^n$$

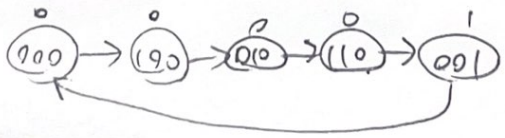
$$Q_3^{n+1} = J_3\overline{Q_3^n} + \overline{K_3}Q_3^n = Q_1^nQ_2^n\overline{Q_3^n}$$

| $Q_1^n$ | $Q_2^n$ | $Q_3^n$ | $Q_1^{n+1}$ | $Q_2^{n+1}$ | $Q_3^{n+1}$ | CO |
|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 1 | 0 | 0 | 0 |
| 0 | 0 | 1 | 0 | 0 | 0 | 1 |
| 0 | 1 | 0 | 1 | 1 | 0 | 0 |
| 0 | 1 | 1 | 0 | 1 | 0 | 1 |
| 1 | 0 | 0 | 0 | 1 | 0 | 0 |
| 1 | 0 | 1 | 1 | 1 | 0 | 1 |
| 1 | 1 | 0 | 0 | 0 | 1 | 0 |
| 1 | 1 | 1 | 1 | 0 | 0 | |




可以自启动. ~~变001. CO输出1~~ 自锁 强制



2. 输出方程：$CO = \overline{\overline{Q_3 Q_2 Q_1 Q_0}} = \overline{Q_3}\,\overline{Q_2}\,\overline{Q_1}\,\overline{Q_0}$

驱动方程：$J_1 = \overline{\overline{Q_2}\,\overline{Q_3}} = Q_2 + Q_3$

$J_3 = \overline{Q_1}\,\overline{Q_2}$

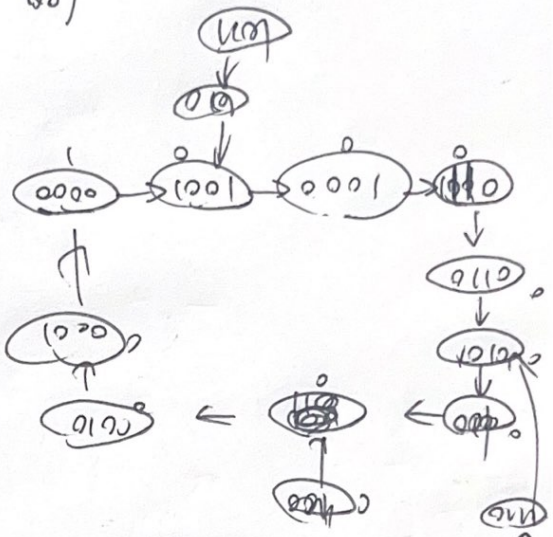
状态方程，$Q_0^{n+1} = \overline{Q_0^n}$

$Q_1^{n+1} = (Q_2^n + Q_3^n)\,\overline{Q_1^n}$

$Q_2^{n+1} = \overline{Q_2^n} \cdot Q_1^n\uparrow$

$Q_3^{n+1} = \overline{Q_3^n}\,Q_2^n\,\overline{Q_3^n} \cdot Q_0^n\uparrow$

| $Q_0^n Q_1^n Q_2^n Q_3^n$ | $Q_0^{n+1} Q_1^{n+1} Q_2^{n+1} Q_3^{n+1}$ | CO |
|---|---|---|
| 0000 | 1001 | 1 |
| 0001 | 1010 | |
| 0010 | 1100 | |
| 0011 | 1 | |
| 0100 | 1000 | |
| 0101 | 1001 | |
| 0110 | 1010 | |
| 0111 | 1010 | |
| 1000 | 0 | |
| 1001 | 0001 | |
| 1010 | 0010 | |
| 1011 | 0 | |
| 1100 | 0100 | |
| 1101 | 0101 | |
| 1110 | 0110 | |
| 1111 | 0111 | |




自启动 异步 10进制

注

0101 0010..

3. 1010.　十进制计数

83进制计数.

4.